



Issues Reflected to the Engineering Field of the Age



CONTENT

- Introduction
- Constant Updating of Knowledge Is Necessary
- Privacy Issues & Security Breaches
- Lack of Communication
- Lack of Well-defined Standardization
- Skill and Staff Shortage
- Increase in Computer Crime
- Teamwork Shortage
- Ethical Responsibility



Introduction

- Because of the ever-increasing technological developments and innovations in our age, many issues have been faced in the field of engineering.
- We will look at and explain these issues in this presentation.



Constant Updating of Knowledge Is Necessary

- A new platform, framework, learning environment, programming language or library almost coming up in every day.
- Hence, you must follow the technological innovations in the field of the engineering and update your skills.



Constant Updating of Knowledge Is Necessary

- In other words, as technology continues to grow and expand, you need to keep up.
- Due to the tools/software/application become outdate quickly, you must update their versions and you also must get used to them fast.



Privacy Issues & Security Breaches

- Among the many issues in our age posed by new information and communications technologies is to address individual data privacy.
- Once data is put into a computer it can easily be copied or transmitted. This puts people's personal data at risk.
- Take the necessary security measures for the protection of personal data plays an important role in preventing data theft.



Privacy Issues & Security Breaches

- Another important faced issue regards security.
- Security breaches are on the rise; almost all web application or software contains serious vulnerabilities.
- Attackers can exploit these vulnerabilities and infiltrate computers.
- In this context, to cope with security threats, keeping each layer of your software secure is so significant.



Privacy Issues & Security Breaches

- To sum up this subtopic, digital age allows that vast amounts of data can be collected and stored.
- These can be private information concerning individuals or organizations. It can be very difficult to keep this data safe.
- Just a single breach can mean vast amounts of private information going into the hands of criminals, business rivals, foreign adversaries, or other malicious people.



Lack of Communication

- ▶ Another problem of living in a digital age is poor communication.
- ▶ For example, when an engineering project is to be carried out, lack of communication may lead to slow progress of the works, obtain wrong or inadequate results.



Lack of Well-defined Standardization

- Today, there are many methods and standards used in computer engineering.
- The standards that will use in engineering studying are should be clear, understandable.



Lack of Well-defined Standardization

- For instance, if the standards are not well-defined, engineers could be using any of their own methods.
- This may lead to some issues such as security concerns and performance issues.



Skill and Staff Shortage

- Engineering are among the most in demand jobs in the world.
- It is a fact that the engineering industry is struggling against a widespread skill and staff shortage.
- It can be said that the need for skilled engineers has never been as urgent as in this age.



Increase in Computer Crime

- In this age, the internet become fertile zone for malevolent forces to operate, thanks to its international nature, vast scale, and the relative anonymity.
- To give an example, unauthorized computer entry.
- This crime involves both an invasion of property and privacy and compromises the computer system itself.



Teamwork Shortage

- Working as a team allow that employees to draw from each other's experience and knowledge.
- Teamwork is recognized as an important skill in the field of the engineering.



Teamwork Shortage

- Today, tendency for people to socialize, communicate via digital devices rather than through real-life contact increased.
- This can easily lead to isolation. Hence, this situation pushes people to work individually instead of collaboration work.



Ethical Responsibility

- Because of the increased technology advancements and engineering studying, computers or machines have been taken over jobs that once belonged to people.
- The most current example of this is the concept of artificial intelligence (AI). This is an attempt by computer experts to simulate the components of human intelligence through machines.



Ethical Responsibility

- Although artificial intelligence technology has become a centralized force, it is an engineering product.
- This situation leads to many ethical dilemmas.



Ethical Responsibility

- As an example, who will be responsible for the consequences of an unethical decision made by artificial intelligence?
- Is it artificial intelligence or engineer who designed and programmed its decision mechanism?

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